SESSION 9 – CONFUSION MATRIX & CLASSIFICATION METRICS

🎯 **Topic**

- Confusion Matrix
- Accuracy
- Precision
- Recall
- F1 Score
- Model Evaluation

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1. 🎯 Learning Objectives

By the end of this session, students will be able to:

- Understand **Confusion Matrix structure**
- Calculate **Accuracy, Precision, Recall, F1 Score**
- Interpret classification results correctly
- Understand **false predictions**
- Evaluate model performance (interview level)

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2. 🔁 Quick Recap (Connect Previous Session)

Start class like this:

"In the previous session, we predicted classes using Logistic Regression.
But now the question is — how good is our model?
Today we will learn how to evaluate classification models properly."

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3. 📖 What is Confusion Matrix?

3.1 Definition

A Confusion Matrix is a table used to evaluate classification models by comparing **actual vs predicted values**.

3.2 Simple Meaning (Most Important Line)

"Confusion Matrix tells how many predictions were correct and how many were wrong."

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4. 📊 Structure of Confusion Matrix

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

4.1 Terms Explained

✅ **True Positive (TP)**

- Actual = Positive
- Predicted = Positive

👉 Correct prediction

❌ **False Positive (FP)**

- Actual = Negative
- Predicted = Positive

👉 Wrong (Type 1 Error)

❌ **False Negative (FN)**

- Actual = Positive
- Predicted = Negative

👉 Wrong (Type 2 Error)

✅ **True Negative (TN)**

- Actual = Negative
- Predicted = Negative

👉 Correct prediction

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5. 📊 Real-Life Example

Example: Disease Prediction

Case	Actual	Predicted
1	Disease	Disease → TP
2	No Disease	Disease → FP
3	Disease	No Disease → FN
4	No Disease	No Disease → TN

Teaching Line

"False Positive means model says yes but actually no,
False Negative means model says no but actually yes."

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6. 📊 Accuracy

Formula

$$Accuracy = \frac{TP + TN}{Total}$$

Meaning

"How many predictions are correct overall."

Problem

👉 Accuracy can be misleading in **imbalanced data**

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7. 📊 Precision

Formula

$$Precision = \frac{TP}{TP + FP}$$

Meaning

"Out of all predicted positives, how many are actually correct?"

Use Case

- Important when **false positives are costly**

Example:

- Spam detection

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8. 📊 Recall (Sensitivity)

Formula

$$Recall = \frac{TP}{TP + FN}$$

Meaning

"Out of all actual positives, how many did we correctly identify?"

Use Case

- Important when **false negatives are dangerous**

Example:

- Disease detection

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9. 📊 F1 Score

Formula

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Meaning

"Balance between precision and recall."

Use Case

- When dataset is imbalanced
- When both FP and FN matter

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10. 📊 Summary Table

Metric	Focus
Accuracy	Overall correctness
Precision	Correctness of positive predictions
Recall	Coverage of actual positives
F1 Score	Balance of precision & recall

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11. 🎯 Interview Concepts

Q1: What is Confusion Matrix?

👉 Table comparing actual vs predicted values.

Q2: Difference between Precision & Recall?

👉 Precision → correctness of predicted positives
👉 Recall → coverage of actual positives

Q3: When Accuracy fails?

👉 In imbalanced datasets

Q4: Why F1 Score?

👉 To balance precision and recall

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12. 💻 PRACTICAL (YOUR FORMAT)

🎯 Objective

Evaluate Logistic Regression model using:

- Confusion Matrix
- Accuracy
- Precision
- Recall
- F1 Score

Output Interpretation

Explain:

Confusion Matrix:

[[TN FP]
 [FN TP]]

Example:

- High TP → good model
- High FP → low precision
- High FN → low recall

👉 Conclusion:

"Different metrics give different insights about model performance."

Accuracy alone is not enough to evaluate a classification model.
We must use precision, recall, and F1 score to understand the model deeply.
A good data scientist always looks beyond accuracy.